From 1965–2025 Barrels Outrun Mass-Based Energy Equivalents, But Both Keep Growing

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World oil production reaches new highs both in barrels and in a mass-based energy equivalent. The equivalent grows more slowly than barrel volume, but the decline per barrel has not stopped aggregate growth.

I. MAIN FINDING AND SCOPE

Global oil production sets a new record in 2025 in both volume and reported mass expressed as a standardized energy equivalent. Using the 2026 Energy Institute (EI) edition for every year, volume rises 5.88% and the daily equivalent 4.70% above 2018. The equivalent already exceeds 2018 in 2023 and 2024; 2025 extends the record to 100.59 million barrels per day (Mb/d) and 0.53925 exajoules per day (EJ/day). We compare with 2018 because it has the largest gross and net-equivalent daily rates within the common production-and-return window, 1971–2020. It is not the last volumetric record. The central finding is direct: slower growth in energy equivalent per barrel has not prevented aggregate output from reaching new highs.

Scope and assumptions. The mass-based equivalent assigns a fixed energy content to each reported tonne. It tracks mass rather than independently measured global heating values. The investment adjustment uses annual joint oil-and-gas returns through 2020 and an explicit allocation to oil. These choices support a reproducible test of peak claims and quantitative thresholds for the missing years; they do not supply the missing oil-specific measurements. Extraction, export capacity, and availability to particular markets also differ, especially during war.

We test whether changes in stream mix and production-energy requirements have offset output growth. The evidence comprises a complete annual table, observed shares of natural gas liquids (NGLs), and explicit break-even conditions. The analytical schematic illustrates how a net-energy peak can occur.

II. SOURCES, CONVERSIONS, AND TIME BASIS

A. Production and source verification

The production inputs are the EI 2026 narrow-format dataset, selecting `Total World` and the variables `oilprod_kbd`, `oilprod_mt`, `oilprod_crudecond_kbd`, and `oilprod_ngl_kbd`, for 1965–2025 [1]. The EI Excel workbook provides a separate cross-check of available overlapping entries. It contains an oil-consumption exajoule series, but no corresponding oil-production exajoule series; Eq. (1) is our conversion of the production mass series. Oil comprises crude oil, tight oil, oil sands, condensates, and gas-plant liquids; biofuels, synthetic liquids from other sources, and refinery processing gain are excluded [2]. Processing gain reconciles volume expansion during refining and contributes no independent extracted energy.

Appendix A gives every annual value. The reproducibility archive retains the source files and their cryptographic checksums. All years are recalculated from EI 2026, rather than appending 2025 to an older series. A separate comparison with the 2025 release [3, 4] quantifies the effect of revisions. Its versioned public mirror preserves that earlier extract; it is an access route to EI data.

B. Energy equivalent and daily rates

Let q_y be reported volume in million barrels per day (Mb/d), M_y annual mass in million tonnes (Mt), and D_y the actual number of days in year y. Define

$$G_y^{\mathrm{std}} = 0.041868 M_y \quad \mathrm{EJ}, \qquad g_y^{\mathrm{std}} = G_y^{\mathrm{std}}/D_y \quad \mathrm{EJ/day}. \tag{1}$$

Equation (1) defines a standardized mass-based estimate rather than measured global heat content. Here EJ means exajoule (10^{18} joules). The factor assigns 41.868 gigajoules (GJ) to each reported tonne. The International Energy Agency (IEA) defines one tonne of oil equivalent (toe) as 41.868 GJ and uses net calorific values in energy balances [5, 6]. This is a lower-heating-value convention; assigning it to each physical tonne is an assumption, not a claim that one tonne of every oil stream equals one toe. EI's conversion appendix likewise specifies lower heating values [7].

Throughout, *gross energy* means before subtraction of production-energy inputs. It does not mean higher heating value (HHV). Lower heating value (LHV) excludes heat recovered by condensing combustion water, whereas HHV includes it. The word *net* in net calorific value concerns that heat, while net supply below concerns production inputs.

The equivalent per barrel is

$$h_y^{\mathrm{std}} = \frac{1000 G_y^{\mathrm{std}}}{q_y D_y} \quad \mathrm{GJ/barrel}. \tag{2}$$

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The unit factor is $10^9/10^6 = 10^3$: one EJ per million barrels equals 10^3 GJ/barrel. Peak comparisons use daily rates to remove leap-year effects; annual totals remain in the appendix. Relative plots use $100X_y/X_{1971}$. Curves interpolate annual observations with a piecewise cubic Hermite interpolating polynomial (PCHIP); all extrema and quoted results use the unsmoothed annual rows.

C. What mass and volume can establish

The per-barrel equivalent in Eq. (2) is proportional to tonnes per barrel. Its decline therefore restates the reported mass/volume relationship; it does not independently identify a decline in heating value. Between 1971 and 2025, barrels per tonne rise from approximately 7.43 to 7.81 and the assigned equivalent per barrel falls by 4.82%.

Let k_y be the actual worldwide mean mass-specific heating value on a consistent LHV basis, and $k_0 = 41.868$ GJ/t the imposed conversion. The chemical-energy rate implied by the reported mass would be

$$e_y^{\mathrm{chem}} = g_y^{\mathrm{std}} \frac{k_y}{k_0}. \tag{3}$$

From Eq. (3), 2025 exceeds 2018 in chemical-energy rate if $k_{2025}/k_{2018} > 0.9551$. Reversing that ranking therefore requires a fall exceeding 4.49% in mean heating value per tonne. The reported decline per barrel does not establish such a fall: it combines tonnes per barrel with energy per tonne. The source supplies no measured global trajectory for k_y. A record against every historical year additionally requires each corresponding comparison.

Reported mass can inherit volume-to-mass conversion assumptions. EI does not document a universal procedure reconstructing every national tonnage series from barrels with fixed national factors. Nor are aggregate national ratios fixed: United States barrels per tonne rise from 8.152 in 2005 to 8.640 in 2025 (archived source-reconciliation record). Such changes can reflect stream mix, country mix, conversion conventions, and revisions; even fixed component factors can produce a changing aggregate. The mass/volume trend therefore establishes the reported relationship without uniquely identifying within-stream quality changes.

The constant factor can also understate a composition effect. EI's illustrative lower heating values are 49.40 GJ/t for ethane and 46.15 GJ/t for liquefied petroleum gas (LPG), both above 41.868 GJ/t [7]. Holding other characteristics fixed, a rising mass share of these streams raises energy per tonne. The fixed-factor calculation then understates energy growth and overstates the per-barrel decline. Quantifying the global correction requires component masses and heating values.

TABLE I. Reported world stream volumes in Mb/d and the NGL share. All years use EI 2026.

Year	Crude + condensate	NGL	Share (%)
1971	48.46	2.32	4.58
2005	74.97	6.83	8.35
2018	83.60	11.40	12.00
2020	77.06	11.88	13.36
2024	82.92	14.23	14.65
2025	85.76	14.83	14.74

III. COMPOSITION EVIDENCE AND SENSITIVITY

For non-overlapping stream flows V_i and heating values per volume h_i, let

$$V = \sum_i V_i, \quad s_i = V_i/V, \quad G = V\bar{h}, \quad \bar{h} = \sum_i s_i h_i. \tag{4}$$

Then

$$\frac{\dot{G}}{G} = \frac{\dot{V}}{V} + \frac{\dot{\bar{h}}}{\bar{h}}, \qquad \dot{\bar{h}} = \sum_i \dot{s}_i h_i + \sum_i s_i \dot{h}_i. \tag{5}$$

Here G denotes actual chemical-energy flow in an analytical identity; it is distinct from the standardized statistical equivalent G_y^{std}. The first sum changes stream weights; the second changes within-stream quality. EI supplies the crude-plus-condensate and NGL volume split, permitting an empirical application of the first component's shares without inventing the second component.

Table I lists selected stream volumes, and Fig. 1 shows the annual composition trajectory. The NGL share grows from 4.58% in 1971 to 12.00% in 2018 and 14.74% in 2025. In the two-stream approximation with fixed per-barrel factors h_C, h_L, the share effect is exactly

$$\Delta\bar{h}_{\mathrm{share}} = (s_{L,b} - s_{L,a})(h_L - h_C). \tag{6}$$

For 2018–2025, the observed share increment is 0.02745, so the composition term is $0.02745(h_L - h_C)$. An illustrative ratio $h_L/h_C = 0.70$ lowers the mixture's per-barrel value by about 0.85% relative to 2018. For context, the United States Energy Information Administration (EIA) lists higher-heating-value factors of 3.841 and 4.353 million British thermal units (Btu) per barrel for propane and normal butane [8]. Dividing by an illustrative crude reference of 5.8 million Btu per barrel gives 0.66 and 0.75. Thus 0.70 represents a plausible propane–butane-rich mixture, rather than an estimated world NGL average. The example explains the share mechanism; it is neither substituted into the LHV series nor added as a second correction to the mass-based series, which already reflects part of the changing mix.

Small margins and source revisions. Table II reports edition-specific annual totals and daily rates; the following comparisons distinguish leap-year effects from revisions. In the 2025 edition, the 2024 daily equivalent exceeds 2018 by 0.876% and 2023 by 0.352%; annual totals give 1.153% and 0.627%. The 2026 edition revises

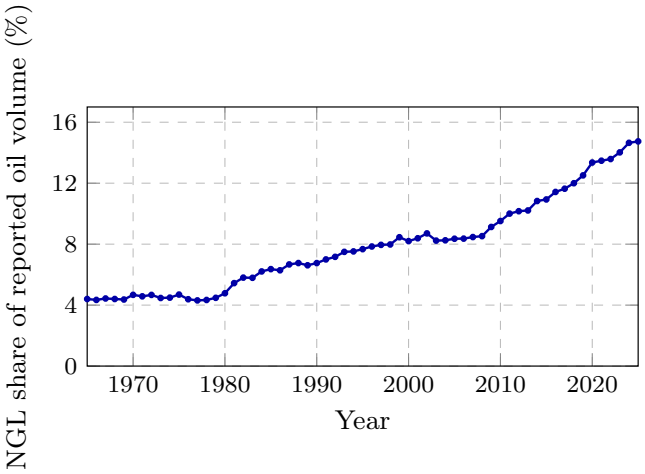


FIG. 1. Observed NGL share of world oil volume (EI 2026). Dots are annual observations; connecting segments guide the eye.

TABLE II. Daily-rate and vintage comparisons. Growth rates use daily equivalents; each comparison uses one edition for both years.

Edition	Year	Annual (EJ)	Daily (EJ/day)	Above 2018 (%)	Above prev. year (%)
2025	2018	188.02	0.51514	0.000	2.394
2025	2023	189.01	0.51783	0.522	1.979
2025	2024	190.19	0.51965	0.876	0.352
2026	2018	187.98	0.51502	0.000	2.365
2026	2023	189.30	0.51863	0.701	1.951
2026	2024	190.72	0.52109	1.178	0.474
2026	2025	196.82	0.53925	4.704	3.485

those daily margins to 1.178% and 0.474%. Its 2025 daily value exceeds 2018 by 4.704% and revised 2024 by 3.485%.

A source revision can reverse a close ranking. If the later estimate is $1+m$ times the earlier one, a downward revision of the later value alone exceeding $m/(1+m)$ reverses it. This threshold is 1.16% for revised 2024 versus 2018 and 4.49% for 2025 versus 2018. The latest record has a substantially larger margin. These are sensitivity thresholds; no statistical error distribution is supplied with the derived global equivalents.

IV. ENERGY-RETURN ACCOUNTING AND THE INVESTMENT-ADJUSTED EQUIVALENT

A. Physical accounting and the conditional oil proxy

Energy return on investment (EROI), R, is returned energy divided by invested energy. For consistently defined flows,

$$R = G/I, \qquad I = G/R, \qquad N = G - I = G(1 - 1/R). \tag{7}$$

G, I, N denote gross returned, invested, and net energy under a consistently specified accounting boundary. For additive component outputs and allocated inputs, $1/R = \sum_i w_i/R_i$, where $w_i = G_i/\sum_j G_j$. Aggregate EROI is thus an output-energy-weighted harmonic mean; barrel-weighted averages of the ratios are inappropriate. Joint oil-and-gas extraction requires allocation of shared inputs.

Long-run studies establish why accounting boundaries matter. Court and Fizaine reconstruct annual fossil-fuel returns through 2012 using a price-based method [9]. Hall, Lambert, and Balogh review declining returns and their societal implications, including the possibility that joint oil-and-gas results mask lower oil returns [10]. Their minimum useful returns concern extended supply boundaries and societal requirements, rather than a discontinuity in Eq. (7). These studies motivate our sensitivities; their different return series are not spliced into the selected annual data.

We use Aramendia and colleagues' annual primary-stage oil-and-gas returns for 1971–2020, including direct and indirect requirements [11, 12]. The exact `global_erois.csv` selection is world (`WRLD`), `Oil and gas products`, `Primary`, and indirect energy `Included`. No world oil-only primary-stage row is available; oil-specific rows cover later stages. Applying the joint return to oil gives

$$\widetilde{i}_y^{\,\mathrm{oil}} = g_y^{\mathrm{std}}/R_y^{OG}, \qquad \widetilde{n}_y^{\,\mathrm{oil}} = g_y^{\mathrm{std}} - \widetilde{i}_y^{\,\mathrm{oil}}. \tag{8}$$

The superscript OG denotes joint oil and gas. We call $\widetilde{n}_y^{\,\mathrm{oil}}$ the *net-equivalent proxy*. Its baseline allocation assigns oil the joint system's input energy per unit of output energy. Higher actual oil intensity would lower the proxy, and lower intensity would raise it. The source includes indirect supply-chain requirements, uses energy inputs measured at the final stage, and omits capital-formation energy. These are annual-flow, primary-stage returns. Later-stage ratios cannot be inserted into this primary-output calculation; boundary differences across energy-return studies matter [13, 14].

The proxy measures the stated investment adjustment at extraction. Refining, distribution, end-use conversion, and regional access require their own compatible accounts.

Table III displays the subtraction and the direct-only alternative. Joint EROI falls from 41.99 in 1971 to 20.90 in 2007, then stays near 19–22 through 2020. The assumed input deduction rises from 2.38% to 4.79% between the endpoints, with annual reversals. Even across the full series the retained fraction remains approximately 94.8–97.6%. The return adjustment lowers the level and modestly reduces growth; it does not turn the long-run rise into a decline. Gross and net-equivalent daily rates both attain their common-window maximum in 2018.

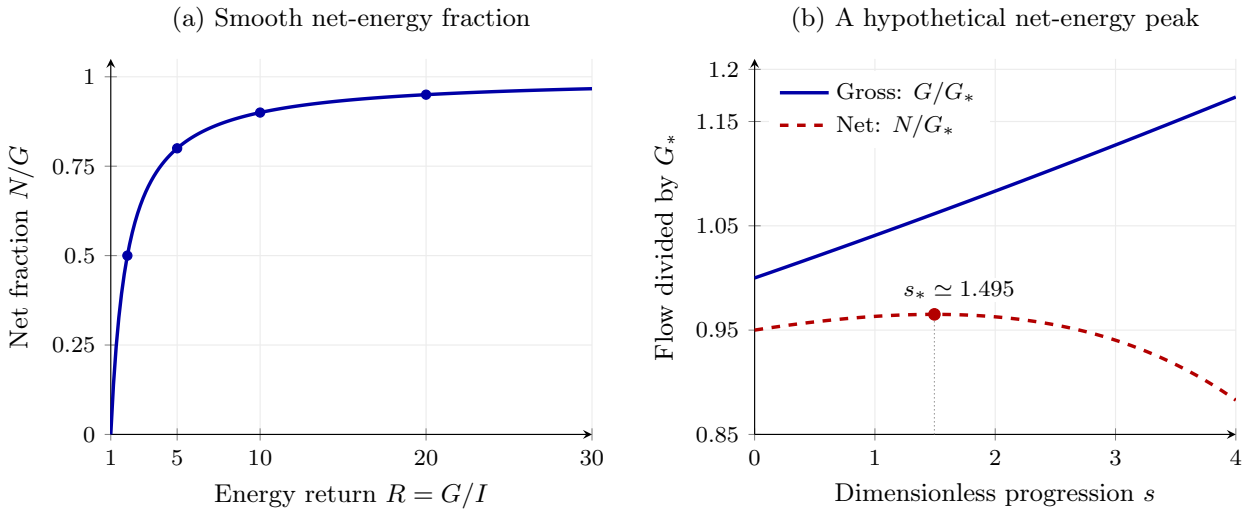


FIG. 2. Analytical net-energy schematics. (a) The retained fraction $N/G = 1-1/R$ changes smoothly with EROI. (b) Growing gross flow can coexist with falling net flow when return declines sufficiently fast. The stated exponential example peaks at $s_* \simeq 1.495$; it is not a historical estimate.

TABLE III. Investment subtraction and input coverage. Energy columns are standardized equivalents in EJ/day. Direct-only excludes indirect requirements.

Year	Gross equiv.	Joint EROI	Deduction	Net incl. indirect	Direct-only net
1971	0.28604	41.99	0.00681	0.27922	0.28095
2005	0.45097	21.93	0.02056	0.43040	0.43320
2013	0.47303	19.19	0.02465	0.44838	0.45221
2018	0.51502	21.03	0.02449	0.49053	0.49364
2019	0.51463	21.01	0.02450	0.49013	0.49324
2020	0.47933	20.86	0.02298	0.45635	0.45925

B. How fast must returns fall?

For differentiable positive gross flow and $R > 1$,

$$\frac{\dot{N}}{N} = \frac{\dot{G}}{G} + \frac{\dot{R}}{R(R-1)}. \tag{9}$$

This follows by differentiating $\ln N = \ln G + \ln(R-1) - \ln R$. If gross energy is increasing and EROI is declining, net energy falls only when

$$-\frac{\dot{R}}{R} > (R-1)\frac{\dot{G}}{G}. \tag{10}$$

In plain terms, the fractional loss rate of EROI must exceed $(R-1)$ times the gross growth rate. At $R \simeq 21$, it must exceed approximately twenty times that rate. The net fraction steepens continuously as return falls: $d(N/G)/dR = 1/R^2$. This makes losses at low return increasingly consequential without imposing a universal threshold.

The historical endpoint comparison is far from the schematic peak regime: between 2005 and 2019, joint EROI falls 4.21% while the daily gross equivalent rises 14.12%. The ratio of the magnitudes of their average logarithmic annual rates is about 0.33, against roughly twenty required locally near $R = 21$. This comparison does not test every intervening year, but it decisively rejects the inference that a falling return by itself entails a falling aggregate net flow.

C. Two analytical schematics

Figure 2(a) shows the net fraction as a function of return; panel (b) illustrates a net-energy maximum alongside growing gross output. The example uses a dimensionless progression s,

$$G(s) = G_* e^{gs}, \qquad R(s) = R_* e^{-ks}, \tag{11}$$

with $g, k > 0$ and $R > 1$. Its stationary point is

$$s_* = \frac{1}{k}\ln\left(\frac{gR_*}{g+k}\right), \qquad R(s_*) = 1 + k/g. \tag{12}$$

For $g = 0.04$, $k = 0.40$, and $R_* = 20$, the strict maximum occurs at $s_* = 1.495$ and $R = 11$. Here $k/g = 10$ and return falls to about 4.04 by $s = 4$. This intentionally strong decline illustrates the mechanism; it is not a calibration to the observed return series. Research models with other production and return assumptions, including Delannoy and colleagues' long-term scenario, are likewise distinct from annual observations [15].

V. HISTORICAL COMPARISONS AND MISSING-YEAR THRESHOLDS

Figure 3 compares indexed daily totals and per-barrel equivalents, while Fig. 4 displays the corresponding absolute daily rates. From 1971 to 2025, volume rises 98.07%

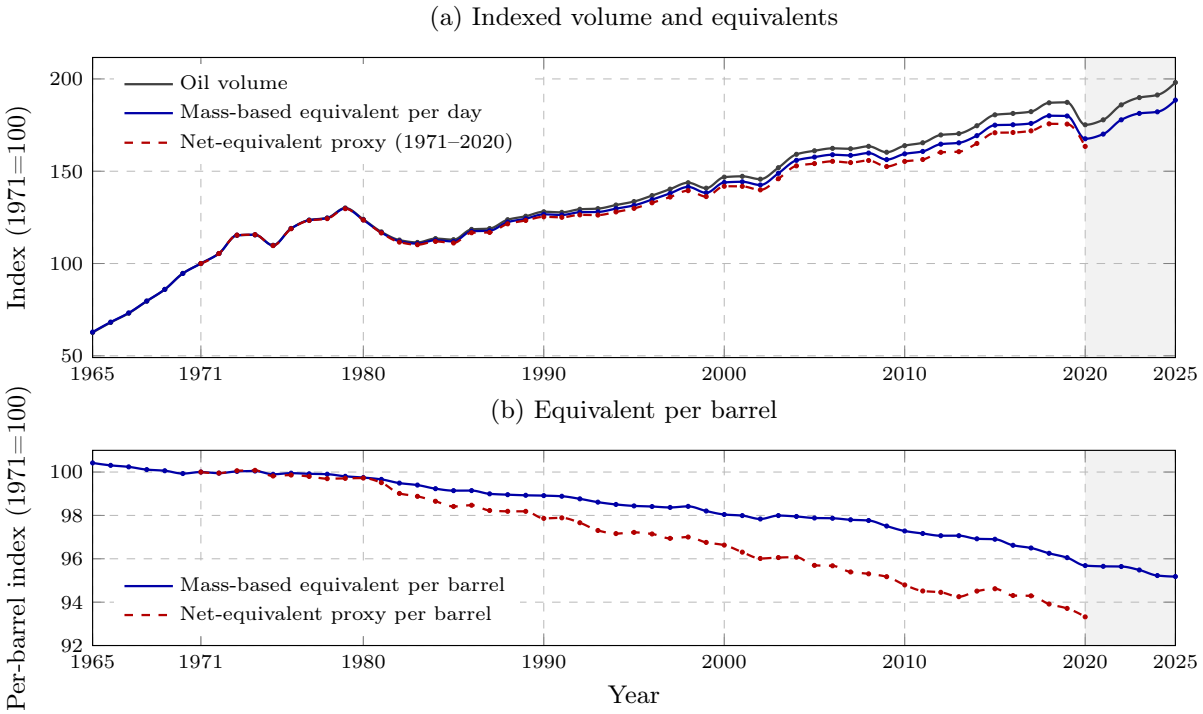


FIG. 3. World oil volume and standardized equivalents (EI 2026; 1971=100). (a) Daily totals, each indexed to its own 1971 value. (b) Per-barrel equivalents, similarly indexed. The net proxy uses Eq. (8) and joint oil-and-gas returns. Gray shading marks 2021–2025, for which returns are unavailable. Dots show annual values; curves are PCHIP interpolation.

and the daily gross equivalent 88.52%. The standardized equivalent per barrel falls 4.82%. Over the common 1971–2020 window, the net-equivalent proxy grows more slowly but remains substantially above its initial level. Neither the growing spread nor the falling per-barrel series implies a long-run fall in aggregate output. Historical contractions remain visible, including 1979–1983 and the 2020 pandemic downturn.

The daily net-equivalent proxy reaches 0.49053 EJ/day in 2018 and 0.49013 in 2019, only 0.082% apart. This is a numerical maximum in the chosen series, insufficient to date a physical oil-only net-energy peak. For scale, the modeled return's 2013–2014 increase from 19.19 to 20.79 raises the retained fraction by 0.423% even at fixed gross output. This annual change exceeds the ranking margin, although it is not itself an uncertainty estimate.

A. A quantitative test of the missing years

For a later year y with a gross equivalent but no published return, the net-equivalent proxy exceeds its 2018 daily value if and only if

$$R_y^{OG} > R_y^* \equiv \left(1 - \frac{\widetilde{n}_{2018}^{\mathrm{oil}}}{g_y^{\mathrm{std}}}\right)^{-1}, \tag{13}$$

provided $g_y^{\mathrm{std}} > \widetilde{n}_{2018}^{\mathrm{oil}}$. Applying the 2026 production series gives $R_{2024}^* = 17.06$ and $R_{2025}^* = 11.07$. Equality gives the same daily net proxy; a higher return gives a higher proxy. The latter threshold is about 47% below the last available joint return of 20.86 in 2020. The source's minimum since 2005 is 19.19 in 2013.

Keeping the 2025 proxy at or below 2018 therefore requires a substantial deterioration in return: 11.07 is about 47% below the last published value and well below the entire 2005–2020 range. Holding the 2020 return constant gives a 2025 proxy of 0.51339 EJ/day, 4.66% above 2018. This is a transparent reference scenario, not an imputed observation. Within the stated boundary, the production record makes a continuing net-equivalent peak in 2018 a claim that requires evidence of sharply rising input intensity.

Changing the accounting boundary requires changing both years consistently. Assigning a 2018 return of 6 or 8 gives 2025 break-even returns of 4.90 or 6.09, respectively, with the same gross-equivalent ratio. These boundary scenarios illustrate the calculation; returns from studies with different inputs and outputs cannot be compared directly with 11.07.

B. Sensitivity to the oil allocation

Let α_y multiply the joint input/output intensity assigned to oil. The baseline has $\alpha_y = 1$; values above one represent higher oil intensity. Then

$$\widetilde{n}_y(\alpha_y) = g_y^{\mathrm{std}}\left(1 - \frac{\alpha_y}{R_y^{OG}}\right). \tag{14}$$

For $0 < \alpha_y < R_y^{OG}$ the adjusted equivalent is positive. For a constant α in both comparison years, recompute the

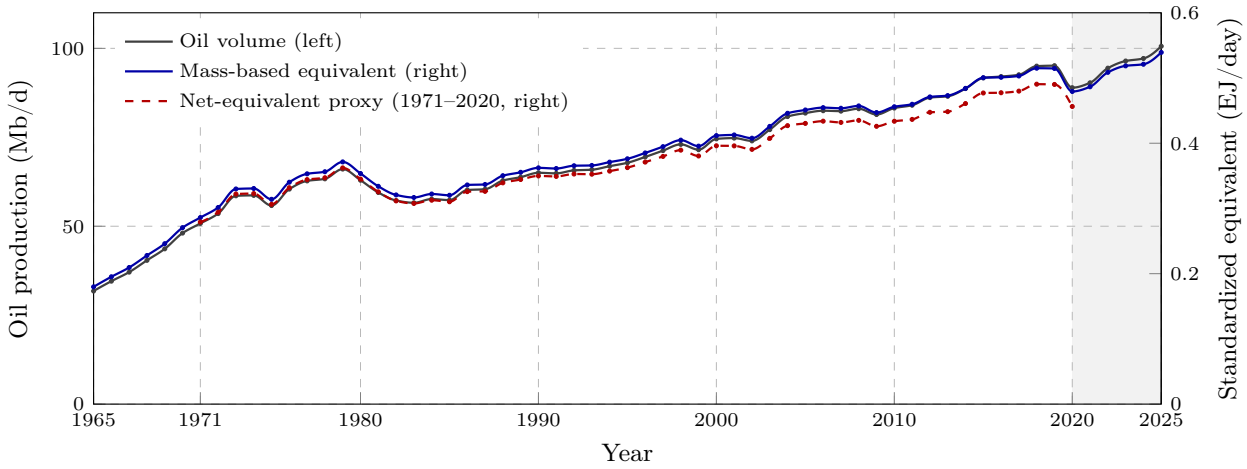


FIG. 4. Absolute daily rates. Volume uses the left axis; standardized equivalents use the right. Both axes begin at zero, with a display ratio of 5.45 GJ per barrel that is not a measured heating value. The gap between energy curves is the investment deduction. Crossings of the volume and energy curves have no physical equality interpretation. The net proxy ends in 2020.

2018 benchmark consistently. The joint return required to exceed it in year y is

$$R_y^*(\alpha) = \frac{\alpha}{1 - \widetilde{n}_{2018}(\alpha)/g_y^{\mathrm{std}}}. \tag{15}$$

For 2025, constant multipliers $\alpha = 0.75$, 1, and 1.25 give thresholds of 9.50, 11.07, and 12.29. The qualitative result survives these substantial changes in allocation: each threshold remains well below the published 2005–2020 joint-return range. These multipliers are stress tests rather than estimated shares or an uncertainty interval. A time-varying multiplier can change the outcome and requires evidence on oil-specific intensity. Full joint-system reconciliation also requires gas output and consistent allocation of shared inputs.

VI. THE 2026 SUPPLY SHOCK AND THE HISTORICAL CUTOFF

The annual production table ends in 2025, before the 2026 Gulf war. It already includes the effects of the Russia–Ukraine conflict and the conditions prevailing in each historical year.

The IEA's October 2026 report documents disrupted Gulf flows and damage affecting Russian refining and exports [16]; September had already reported continuing Gulf shut-ins and reduced Russian product exports [17]. These observations concern current availability. They do not provide a complete 2026 annual mass series with matching returns and are not spliced into the figures.

War can interrupt production and deliveries while recoverable resources remain. Such losses matter immediately, but identifying a structural resource-energy peak requires distinguishing disruption from depletion, changing inputs, policy, and demand. The record through 2025 therefore establishes the historical result without predicting uninterrupted growth or secure access thereafter.

VII. CONCLUSIONS

World oil output reaches new highs in 2025 in both barrels and reported mass expressed as a standardized energy equivalent. Compared with 2018, daily volume rises 5.88% and the equivalent 4.70%. The decline in equivalent per barrel has slowed aggregate growth, not stopped it. The record through 2025 rejects a peak in this mass-based series.

The rising NGL share identifies a real change in the reported stream mix. It does not establish that falling energy per barrel has offset production growth. Reversing the 2025–2018 chemical-energy ranking would require a fall exceeding 4.49% in mean heating value per tonne; the observed volume and mass relationship does not demonstrate that condition.

Declining EROI is likewise insufficient by itself to establish falling net output. With the selected primary-stage return and proportional oil allocation, the 2025 net-equivalent proxy exceeds its 2018 maximum whenever the missing return exceeds 11.07. Maintaining the earlier maximum requires a return about 47% below the last published value. Constant allocation multipliers from 0.75 to 1.25 shift the threshold to 9.50–12.29 without changing that conclusion. This is a strong conditional result, while the actual 2025 oil-only return remains unmeasured here.

The evidence supports continued growth through 2025 in the reported aggregate and sets explicit conditions that an earlier physical or net-energy peak must satisfy. Resolving those remaining claims requires global heating values, documented mass–volume conversions, and updated oil-specific inputs under a consistent boundary. Wartime availability is a separate empirical question.

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DATA AVAILABILITY

The accompanying archive includes the complete 1965–2025 source-selected table, a formula-based workbook, original EI source files, the published energy-return extract, editable TikZ figures, processing programs, a source manifest, and the source-reconciliation record documenting the CSV/Excel checks and national mass–volume ratios. Missing net years remain blank or explicitly unavailable. Source inputs are separated from calculated equivalents and sensitivity scenarios.

ABBREVIATIONS AND NOTATION

Energy and methods. EROI: energy return on investment; **NGL, NGLs**: natural gas liquids; **LPG**: liquefied petroleum gas; **HHV**: higher heating value; **LHV**: lower heating value; **PCHIP**: piecewise cubic Hermite interpolating polynomial.
Organizations. EI: Energy Institute; **IEA**: International Energy Agency; **EIA**: United States Energy Information Administration; **FWF**: Austrian Science Fund (Fonds zur Förderung der wissenschaftlichen Forschung); **TU**: Technische Universität (as in TU Wien).
Data and identifiers. CSV: comma-separated values; **DOI**: digital object identifier; **CC BY**: Creative Commons Attribution; **WRLD**: world identifier in the energy-return dataset.
Units. Btu: British thermal unit; **GJ**: gigajoule (10^9 joules); **EJ**: exajoule (10^{18} joules); **Mt**: million metric tonnes; **t**: metric tonne (1000 kilograms); **Mb/d** and **kb/d**: million and thousand barrels per day; **bbl**: barrel (42 United States gallons); **d**: day; **toe**: tonne of oil equivalent (41.868 GJ).
Symbols. G, I, N, R: gross returned energy, invested energy, net energy, and EROI; $G_y^{\mathrm{std}}, g_y^{\mathrm{std}}$: annual and daily standardized equivalents; $\widetilde{i}_y^{\mathrm{oil}}, \widetilde{n}_y^{\mathrm{oil}}$: daily oil investment deduction and net proxy; R_y^{OG}, R_y^{*}: joint primary return and break-even return (OG denotes oil and gas); k_y, k_0, α_y: actual mean heating value per tonne, imposed conversion, and oil-intensity multiplier; M_y, q_y, D_y: annual mass, daily volume, and days in year y; h_y^{std}: equivalent per barrel.

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Appendix A: Complete annual production and standardized-equivalent table

All production inputs use EI 2026. Energy quantities assign 41.868 GJ per reported tonne. Returns are joint primary-stage oil-and-gas EROI including indirect requirements; the oil proxy assumes the joint intensity and omits capital energy. A dash marks an unavailable return or net proxy. Display precision does not represent measurement uncertainty.

TABLE IV. Annual world oil accounting, 1965–2025. Volume: Mb/d; mass: Mt; annual equivalent: EJ; daily equivalents: EJ/day; per-barrel equivalent: GJ/bbl.

Year	Volume	Mass	Annual equiv.	Daily equiv.	EROI	Daily net proxy	Gross/bbl	NGL (%)
1965	31.792	1567.59	65.63	0.17981	–	–	5.656	4.40
1966	34.561	1702.22	71.27	0.19526	–	–	5.650	4.34
1967	37.104	1826.18	76.46	0.20948	–	–	5.646	4.43
1968	40.418	1992.26	83.41	0.22790	–	–	5.639	4.40
1969	43.649	2144.48	89.79	0.24599	–	–	5.636	4.37
1970	48.075	2358.89	98.76	0.27058	–	–	5.628	4.67
1971	50.785	2493.62	104.40	0.28604	41.99	0.27922	5.632	4.58
1972	53.554	2635.42	110.34	0.30148	42.03	0.29430	5.629	4.67
1973	58.552	2875.76	120.40	0.32987	42.54	0.32212	5.634	4.47
1974	58.671	2882.17	120.67	0.33061	42.53	0.32283	5.635	4.49
1975	55.811	2737.56	114.62	0.31402	40.66	0.30629	5.626	4.69
1976	60.440	2974.27	124.53	0.34024	40.55	0.33185	5.629	4.38
1977	62.740	3078.15	128.88	0.35308	39.97	0.34425	5.628	4.31
1978	63.322	3106.12	130.05	0.35629	38.71	0.34709	5.627	4.34
1979	66.067	3237.58	135.55	0.37137	40.43	0.36219	5.621	4.48
1980	62.942	3091.11	129.42	0.35360	41.60	0.34510	5.618	4.78
1981	59.493	2911.29	121.89	0.33395	39.55	0.32550	5.613	5.44
1982	57.238	2796.11	117.07	0.32073	35.10	0.31160	5.604	5.80
1983	56.592	2762.04	115.64	0.31682	34.58	0.30766	5.598	5.79
1984	57.640	2816.18	117.91	0.32215	33.86	0.31264	5.589	6.21
1985	57.345	2791.55	116.88	0.32021	32.26	0.31028	5.584	6.36
1986	60.176	2929.44	122.65	0.33603	32.83	0.32579	5.584	6.28
1987	60.406	2936.21	122.93	0.33680	31.86	0.32623	5.576	6.67
1988	62.840	3061.70	128.19	0.35024	31.86	0.33925	5.573	6.76
1989	63.792	3098.67	129.73	0.35544	32.13	0.34438	5.572	6.61
1990	65.022	3157.93	132.22	0.36224	29.27	0.34986	5.571	6.76
1991	64.866	3149.42	131.86	0.36126	29.72	0.34911	5.569	7.00
1992	65.704	3195.01	133.77	0.36549	28.80	0.35280	5.563	7.17
1993	65.892	3190.47	133.58	0.36597	27.21	0.35252	5.554	7.49
1994	66.877	3234.67	135.43	0.37104	26.93	0.35726	5.548	7.52
1995	67.841	3279.06	137.29	0.37613	27.84	0.36262	5.544	7.67
1996	69.478	3366.40	140.94	0.38509	27.50	0.37109	5.543	7.85
1997	71.250	3441.38	144.08	0.39475	26.34	0.37976	5.540	7.95
1998	73.014	3528.35	147.72	0.40473	26.42	0.38941	5.543	7.98
1999	71.515	3448.38	144.38	0.39555	26.17	0.38044	5.531	8.45
2000	74.553	3598.81	150.67	0.41168	26.41	0.39609	5.522	8.20
2001	74.774	3597.76	150.63	0.41269	24.63	0.39593	5.519	8.39
2002	73.990	3554.37	148.81	0.40771	23.82	0.39059	5.510	8.71
2003	77.152	3712.22	155.42	0.42582	23.20	0.40746	5.519	8.23
2004	80.829	3898.14	163.21	0.44592	23.52	0.42696	5.517	8.25
2005	81.802	3931.47	164.60	0.45097	21.93	0.43040	5.513	8.35
2006	82.469	3963.03	165.92	0.45459	21.89	0.43382	5.512	8.36
2007	82.351	3954.53	165.57	0.45361	20.90	0.43191	5.508	8.46
2008	83.029	3996.63	167.33	0.45719	20.67	0.43507	5.506	8.53
2009	81.399	3897.26	163.17	0.44704	21.19	0.42595	5.492	9.13
2010	83.211	3974.69	166.41	0.45592	20.50	0.43369	5.479	9.52
2011	84.006	4007.89	167.80	0.45973	19.82	0.43654	5.473	10.00
2012	86.155	4117.40	172.39	0.47100	19.97	0.44742	5.467	10.16
2013	86.525	4123.81	172.66	0.47303	19.19	0.44838	5.467	10.21
2014	88.689	4220.69	176.71	0.48414	20.79	0.46085	5.459	10.83
2015	91.695	4362.81	182.66	0.50044	21.36	0.47702	5.458	10.93
2016	92.076	4380.33	183.40	0.50108	21.18	0.47742	5.442	11.42
2017	92.570	4386.12	183.64	0.50312	21.68	0.47991	5.435	11.64
2018	95.001	4489.87	187.98	0.51502	21.03	0.49053	5.421	12.00
2019	95.127	4486.47	187.84	0.51463	21.01	0.49013	5.410	12.51
2020	88.943	4190.23	175.44	0.47933	20.86	0.45635	5.389	13.36
2021	90.315	4241.79	177.60	0.48656	–	–	5.387	13.47
2022	94.434	4434.82	185.68	0.50870	–	–	5.387	13.58
2023	96.438	4521.34	189.30	0.51863	–	–	5.378	14.02
2024	97.156	4555.21	190.72	0.52109	–	–	5.363	14.65
2025	100.590	4701.08	196.82	0.53925	–	–	5.361	14.74